Statistical Computing

Visualization of Multivariate Data

Spring 2026

Outline

Introduction

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- Visualization of multivariate data is related to exploratory data analysis (EDA) and statistical graphics.
- "Exploratory" is in contrast to "confirmatory" (e.g., hypothesis testing).
- Exploratory work is useful before hypothesis testing to learn appropriate questions and methods.
- With multivariate data, we may also be interested in dimension reduction or finding structure/groups.
- Here we restrict attention to methods for visualizing multivariate data.

Panel Displays

Panel Displays

- A panel display is an array of two-dimensional graphical summaries of pairs of variables.
- A scatterplot matrix displays scatterplots for all pairs of variables in an array.
- Along the diagonal, one may display variable names or other summaries (e.g., densities).
- Before plotting, one may standardize each one-dimensional sample.
- Panel displays can also include arrays of three-dimensional plots.

Figure: Scatterplot Matrix with Diagonal Densities

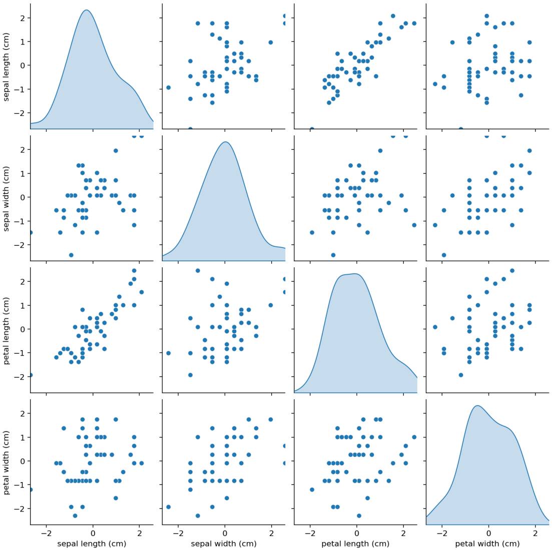
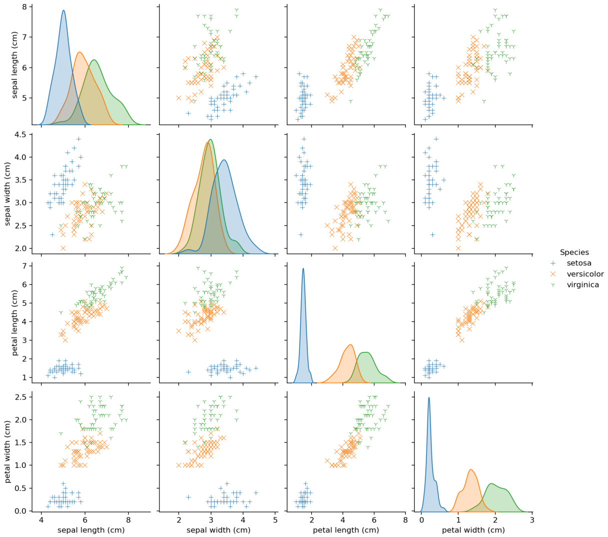


Figure: Scatterplot Matrix by Species

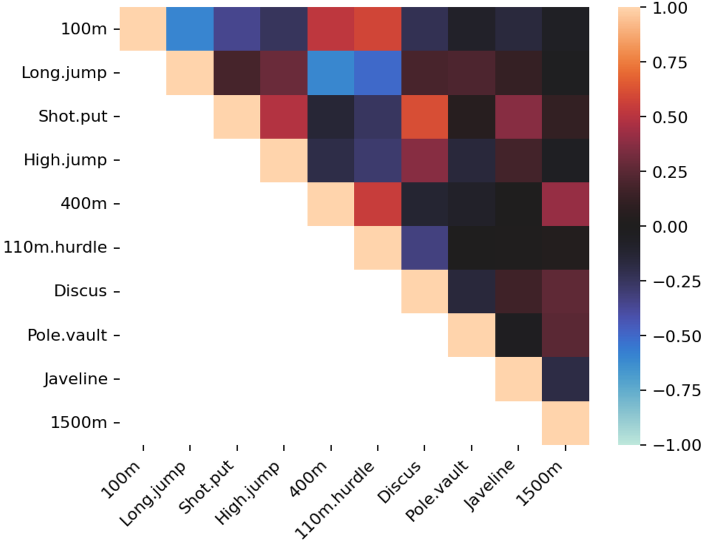


Correlation Plots

Correlation Plots

- With more than about four or five quantitative variables, an array of scatter plots is not helpful for pairwise association.
- A correlation matrix can always be computed, but a large array of correlations is difficult to interpret.
- A graphical summary of pairwise correlations can be very useful.

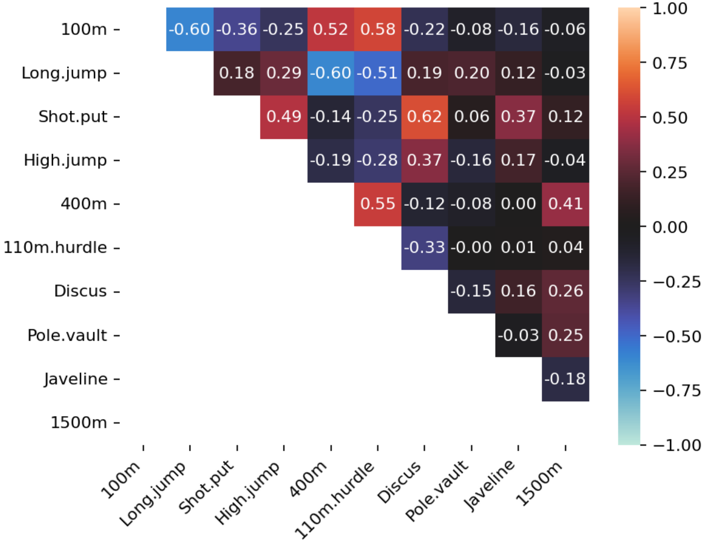
Figure: Correlation Plot of Decathlon Performance



Interpretation

- The correlation plot makes it easy to identify large correlations.
- There are high correlations between pairs of field events like shot put and discus, and between pairs of shorter distance races and hurdles.
- The correlations between pairs of a track event and a field event are small.
- The long jump is negatively correlated with the 400 meter run and 110 meter hurdles.

Figure: Correlation Plot with Correlation Coefficients



Surface Plots and 3D Scatter Plots

- Several tools provide surface and contour plots; a perspective plot draws a surface over the plane.
- For some graphs we mesh a grid of regularly spaced points in the plane.
- To compute function values on a grid, we can evaluate a bivariate function on all (x_i, y_j) pairs.
- We also consider three-dimensional scatterplots to explore possible groups or clusters in data.

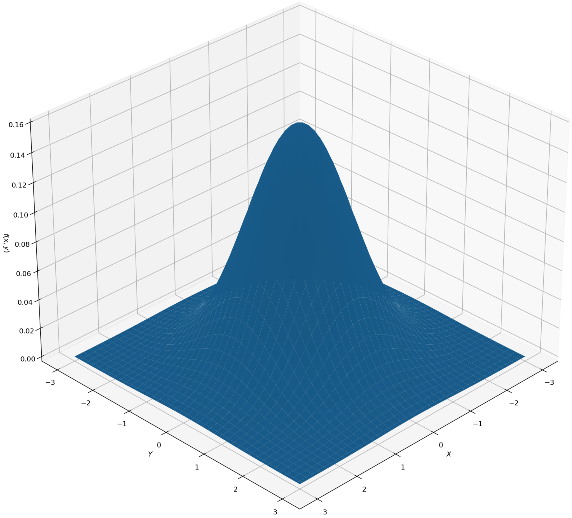
Surface Plots

- Example: plot the standard bivariate normal density

$$f(x,y) = \frac{1}{2\pi} \exp\left(-\tfrac{1}{2}(x^2 + y^2)\right), \quad (x,y) \in \mathbb{R}^2.$$

- We create vectors of x and y values, then compute a full grid of $z_{ij} = f(x_i, y_j)$.

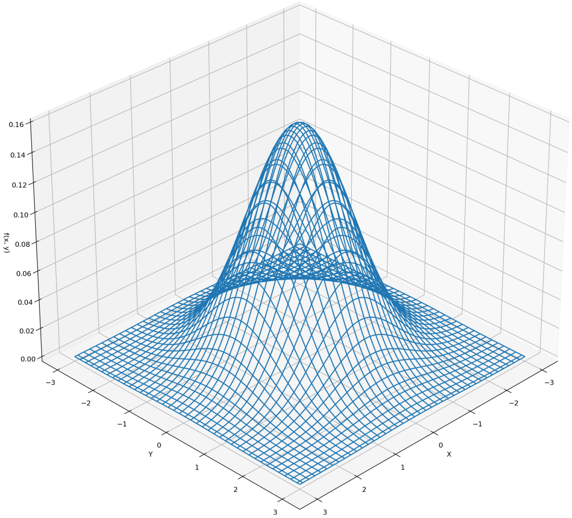
Figure: Perspective Plot of Standard BVN Density



Other Functions for Graphing Surfaces

- Surfaces can also be displayed as a wireframe (mesh) plot.
- Interactive 3D displays allow rotation/tilting to view the surface from different angles.

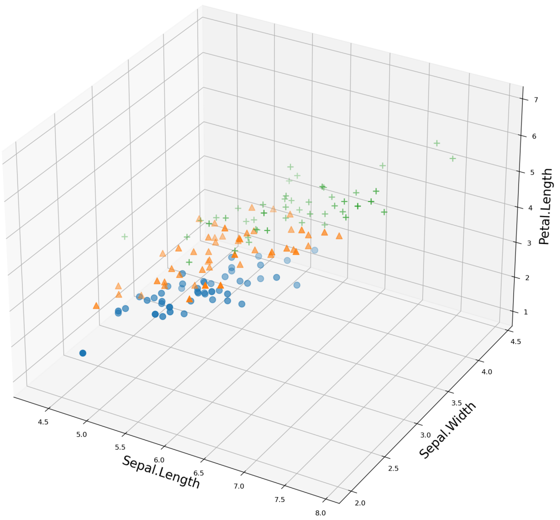
Figure: Wireframe Surface Plot



Three-dimensional Scatterplot

- A 3D scatterplot can help explore whether groups or clusters exist in multivariate data.
- Example: the iris data (three species, four measured variables).
- Plot a 3D scatterplot of three variables with species indicated by plotting symbols.

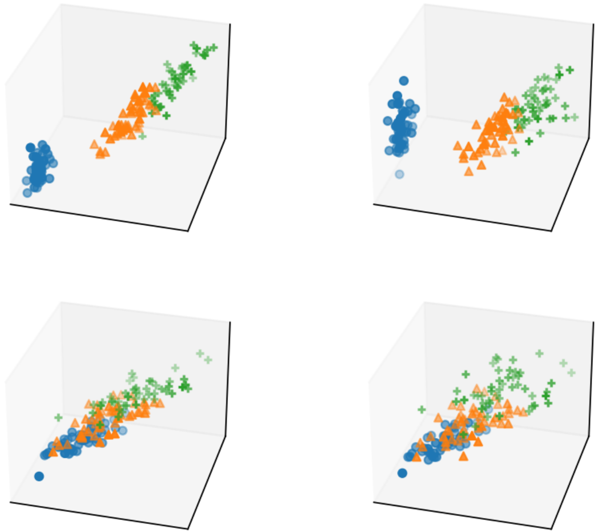
Figure: 3D Scatterplot of Iris Data



Panel Display of 3D Scatterplots

- With four variables, there are four subsets of three variables to graph.
- Arrange four 3D scatterplots in a 2×2 panel display.
- Use a fixed viewing orientation across panels to compare structure.

Figure: Panel Display of Four 3D Scatterplots



Interpretation

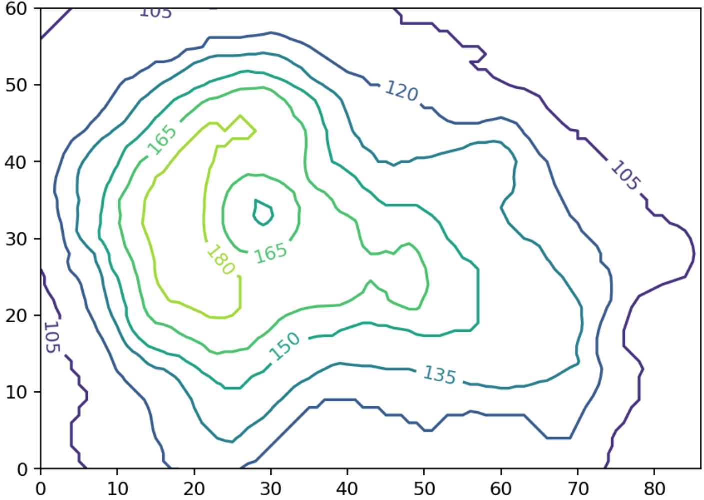
- The three species of iris are separated into groups or clusters in the three-dimensional subspaces spanned by any three of the four variables.
- There is some structure evident in these plots.
- One might follow up with cluster analysis or principal components analysis to analyze the apparent structure in the data.

Contour Plots

Contour Plots

- A contour plot represents a 3D surface $(x, y, f(x, y))$ in the plane by projecting level curves $f(x, y) = c$.
- Contour plots and filled contour plots are common variants; contours are often labeled by default.
- A variation uses color to identify contour levels (image-style display).

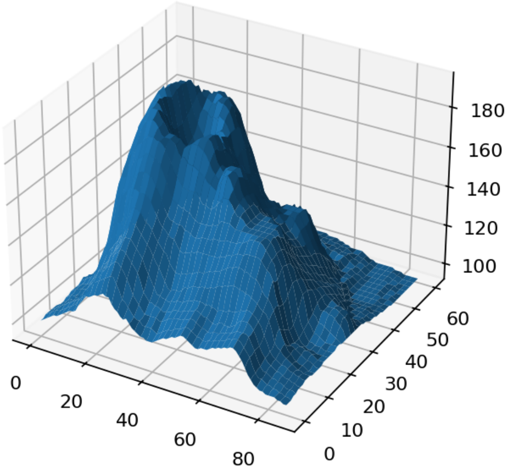
Figure: Contour Plot of Volcano Topography



3D View of the Volcano Surface (for Comparison)

- It can be interesting to compare the contour plot with a 3D surface display of the volcano data.
- An interactive 3D view allows rotation/tilting to view the surface from different angles.
- Another 3D view with shading (to indicate contour levels) can also be produced.

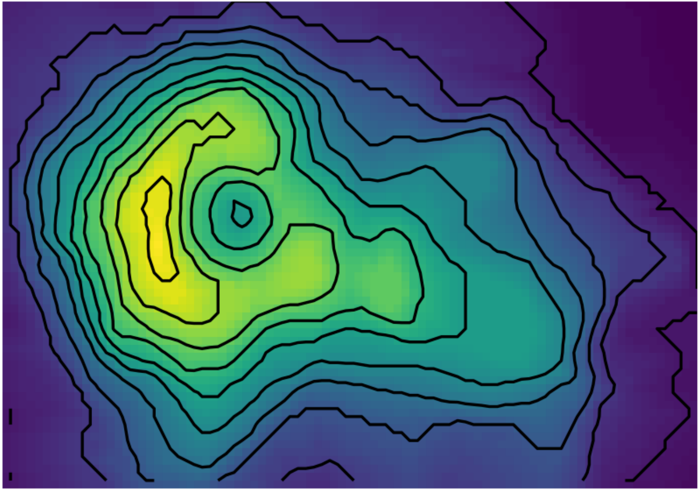
Figure: 3D Surface Plot of Volcano Data



Filled Contour Plots

- A contour plot with a 3D effect can be displayed in 2D by overlaying contour lines on a color map of the height.
- Using an image-style color background without contour lines yields a display similar in spirit to filled contour plots.

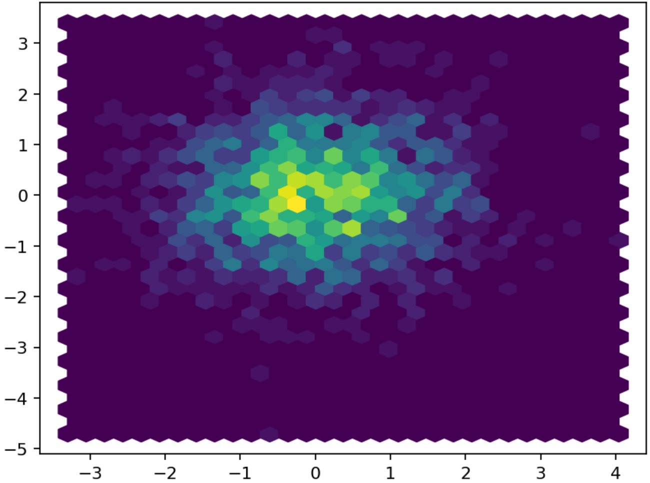
Figure: Image Display with Overlaid Contours



2D Histogram (Flat Density Histogram)

- In large data sets, 2D scatterplots can hide structure when some regions are very dense.
- A 2D (flat) histogram represents the density in each bin using color intensity.
- Example: simulated standard bivariate normal data displayed with hexagonal bins.

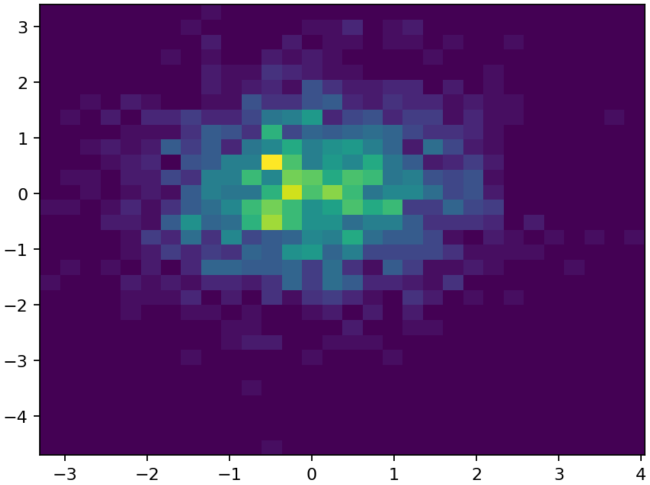
Figure: Flat Density Histogram with Hexagonal Bins



Square-bin Variant (2D Histogram)

- A similar display can use square bins and a color palette to represent density per bin.

Figure: 2D Histogram with Square Bins



Other 2D Representations of Data

Other 2D Representations of Data

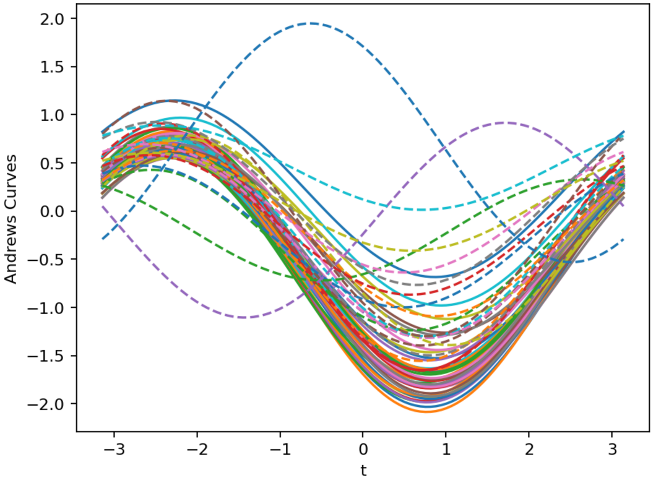
- Beyond contour plots and other 2D projections, there are other methods for representing multivariate data in two dimensions.
- These include Andrews curves, parallel coordinate plots, and iconographic displays (e.g., segment plots, star plots).

Andrews Curves

- For $X_1, \ldots, X_n \in \mathbb{R}^d$, Andrews curves map each observation $x_i = (x_{i1}, \ldots, x_{id})$ to a function $f_i(t)$.
- The mapping uses an orthogonal basis $\{2^{-1/2}, \sin(kt), \cos(kt)\}$.
- Differences are amplified more in lower-frequency terms, so the representation depends on the order of variables/features.

$$f_i(t) = \frac{x_{i1}}{\sqrt{2}} + x_{i2} \sin t + x_{i3} \cos t + x_{i4} \sin 2t + x_{i5} \cos 2t + \cdots, \quad -\pi \leq t \leq \pi.$$

Figure: Andrews Curves by Leaf Architecture



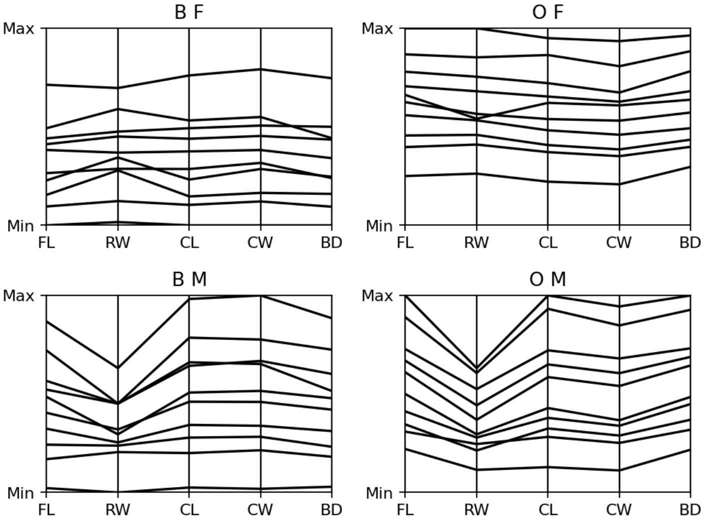
Interpretation

- The plot reveals similarities within the two leaf architecture groups and differences between these groups.
- In general, this type of plot may reveal possible clustering of data.

Parallel Coordinate Plots

- Parallel coordinate plots visualize vectors in $\mathbb{R}^d$ using d equidistant parallel axes.
- Each coordinate is plotted on its corresponding axis, and the points are joined by line segments.
- Axes are often shown as parallel lines (commonly horizontal in this representation).
- Example: a panel display for the crabs data (groups by species and sex), using black and white and a subset for readability.

Figure: Parallel Coordinate Plots by Species and Sex



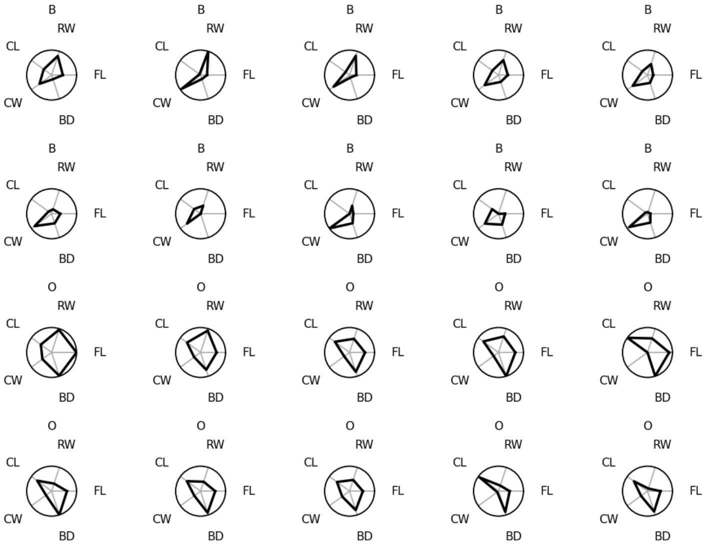
Interpretation

- The axes correspond to the five measurements (frontal lobe size, rear width, carapace length, carapace width, body depth).
- Much of the variability between groups is in overall size.

Segments, Stars, and Other Representations

- Multivariate data can be represented by a two-dimensional icon (glyph), such as a star.
- These icon displays are best shown in a table so observations can be compared.
- They are not very practical for high dimension or large data sets, but can be useful for small data.
- Example: a radar (star) plot for a subset of the crabs data (males), using size-adjusted measurements.

Figure: Segment (Star) Plot for Male Crabs



Interpretation

- Differences between the species (for males) in this sample are evident in the segment plot.
- The plot suggests, for example, that orange crabs have greater body depth relative to carapace width than blue crabs.

Principal Components Analysis

Principal Components Analysis

- PCA is a *dimension reduction* method for multivariate data with many (often correlated) variables.
- It provides an approximate representation of the data in a lower-dimensional space.
- PCA for visualization uses *projections*: project data onto directions that explain the most variation.
- The first principal component (PC1) is the direction of maximal variance; subsequent PCs maximize variance subject to orthogonality.
- If most variation is captured by the first $k < p$ PCs, then $(Z_1, \ldots, Z_k) \in \mathbb{R}^k$ approximates $X \in \mathbb{R}^p$.

PCA via Variance-maximizing Projections

- Let $X = (X_1, \dots, X_p)^\top$, $E[X] = \mu$, $\mathrm{Cov}(X) = \Sigma$.
- Consider $Y = w^\top (X - \mu)$ with $\|w\| = 1$; choose w to maximize $\mathrm{Var}(Y)$.
- Spectral decomposition: $\Sigma = P \Lambda P^\top$, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$.
- PC1: $Z_1 = v_1^\top (X - \mu)$, where v_1 is the eigenvector of λ_1; then $\mathrm{Var}(Z_1) = \lambda_1$.
- PCi: $Z_i = v_i^\top (X - \mu)$, $i = 1, \dots, p$, and the Z_i are uncorrelated.

Basic Properties and Practical Notes

- $\mathrm{Var}(Z_i) = \lambda_i$ and $\mathrm{Var}(Z_1) \geq \cdots \geq \mathrm{Var}(Z_p) \geq 0$.
- $\mathrm{Cor}(Z_i, Z_j) = 0$ for $i \neq j$.
- Fraction of variance captured by first k PCs:

$$\frac{\sum_{i=1}^{k} \lambda_i}{\sum_{j=1}^{p} \lambda_j}.$$

- The coordinates of eigenvectors v_i are called *loadings*.
- PCA is not scale-invariant: if variables are on different scales, scale to common variance before computing PCs.

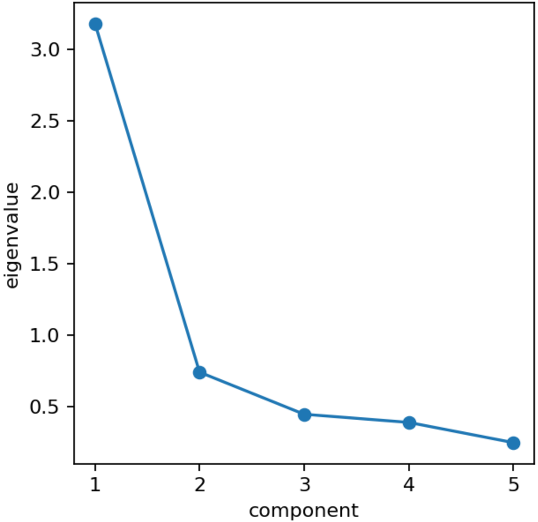
Sample PCA and Covariance Estimators

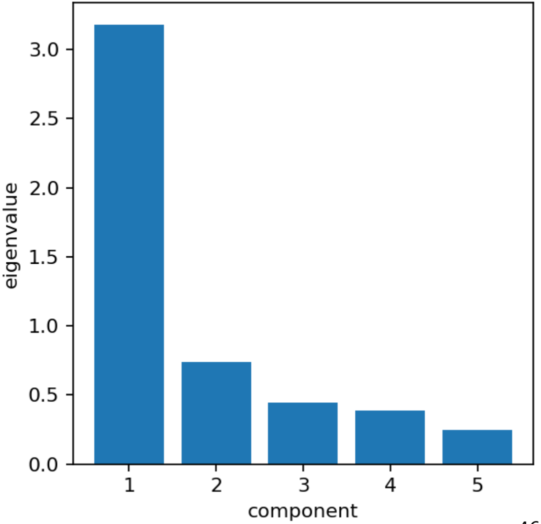
- In practice, replace μ by the sample mean and Σ by a sample covariance matrix.
- Two common estimators: one divides by n (ML estimator), another divides by $n-1$ (unbiased estimator).
- Different software may use different estimators, which can lead to small differences in results.

Example: Exam Scores (Open/Closed Book)

- Data: five exam scores for 88 students (mechanics, vectors, algebra, analysis, statistics).
- Mechanics and vectors are closed book; the other three are open book.
- Pairwise scatterplots/correlations show several highly correlated variables.
- Goal: use PCA to summarize relationships with uncorrelated components and assess whether 2–3 PCs suffice for visualization.

Figure: Scree Plot of Eigenvalues





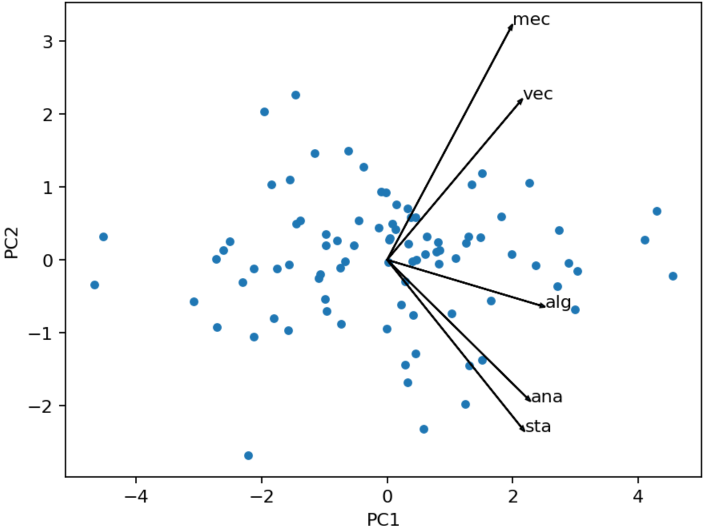
From Eigenvectors to Principal Components

- Let X be the centered and scaled data matrix and P the matrix of standardized eigenvectors.
- The principal component score matrix is $Z = XP$.
- The columns of Z are the PCs (PC1, PC2, ...).

PC Biplot (First Two PCs)

- A PC biplot displays the data in the $(\mathrm{PC1}, \mathrm{PC2})$ coordinate system and adds loading vectors.
- This gives a low-dimensional view for visualization and interpretation of components.

Figure: PC Biplot of Exam Scores



Interpretation from the Biplot and Loadings

- The algebra loading has almost zero slope with respect to PC2, indicating algebra is lightly weighted in PC2.
- Statistics and analysis have positive weight on PC2, while mechanics and vectors have negative weight in the PC2 direction.
- Algebra has its highest loading on PC5 (a component explaining less than 5% of total variance).